

and compact cars. By integrating in a compact package the capability for high-speed and fine-grained processing of large amounts of data from sensors, these make it possible to build systems that are more compact and lower in cost. In addition, the SH74552 is equipped with two-channel FlexRay controller.

The main features of the SH74552 and SH74562 are compact package measuring 13×13 mm², high-speed operation at 160 MHz, 1 MB of on-chip flash memory and 256 kB of on-chip SRAM.

Quad radio chip

Texas Instruments Incorporated (TI) claims to have unveiled the industry's first chip that integrates WLAN 802.11n, GPS, FM transmit/receive and Bluetooth technologies. Integrating



these capabilities on a true single chip, the 65-nanometre WiLink 7.0 solution reduces costs by 30 per cent and size by 50 per cent.

Says Haviv Ilan, vice president and general manager, wireless connectivity solutions, TI, "With its ability to support simultaneous use of all four radios, the WiLink 7.0 solution will truly revolutionise the way people interact with their devices and connect to the larger world."

Chip with field-signal termination and protection

STMicroelectronics claims to have developed the world's first device combining field signal-termination



STMicroelectronics SCLT3-8BT8 device

and protection functions, enabling designers of industrial control and building-automation equipment to increase system performance and EMI robustness while simultaneously cutting power consumption, size and heat dissipation.

ST's SCLT3-8BT8 device allows designers to increase the number of channels in new equipment designs without adding large numbers of extra components. Using conventional terminations, one optical transistor is required per data channel. The SCLT3-8BT8 combines eight data channels into a single serial output, allowing the number of optical transistors and associated components to be reduced significantly. This approach can save in excess of 50 per cent of PC-board area and about 30 per cent of power consumption, compared to existing alternatives. The effective power consumption is less than 78 mW per input.

High-reliability read method for MRAMs

Fujitsu Laboratories and the University of Toronto have jointly developed an innovative circuit design that resolves the problem of erroneous writes in spin-torque-transfer (STT) MRAMs during read operations. This marks a major step towards the practical implementation of STT MRAM as a necessary replacement for flash memory, in view of the future requirements that will be necessary for compact and low-power electronic devices.

The newly developed read method uses a negative resistance that is intermediate between the high resistance of the magnetic tunnel junction (MTJ) and low resistance on a parallel circuit. If the MTJ is in a high-resistance

state, this circuit exhibits negative-resistance characteristics. If the MTJ is in a low-resistance state, then it exhibits normal-resistance characteristics. These characteristics allow the resistance value to be read at lower voltages than before, suppressing the tendency

of the read operation to reverse the direction of magnetization and avoiding the problem of erroneous write operations.

It is anticipated that the STT MRAM used as miniaturised non-volatile memory would enable greater performance in mobile phones and other electronic devices.

Solidstate drive in a chip

Japan's Keio University claims to have developed, in conjunction with Toshiba, the technology to shrink the size of solid-state disks (SSDs) by a spectacular 90 per cent while also cutting energy consumption by 70 per cent. So the entire SSD can be put in the footprint of a single chip.

The researchers used inductive coupling to link a stack of 128 NAND flash chips and a controller. So a single controller can talk to any of the flash chips in the 128-chip stack. The final product of this partnership is expected to yield 1TB drives with 2Gbps throughput. The NAND flash chips atop one another communicate using short-range magnetic signals, which results in less wiring complexity and reduced production costs.

World's first 30nm DRAM chip

Samsung claims to have developed the world's first 30-nanometre dynamic random-access memory (DRAM) chip, which is said to lower power consumption to 85 per cent of existing 40-nanometre DRAM chips.

Based on nanotechnology, it will be the most efficient DRAM solution that can be introduced in consumer electronics and server systems, says the company. The productivity can also be improved by as much as 60 per cent, as